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HamidReza Saadi Dadmarzi

Education

Koç University, Istanbul, Turkey *2023/11 – 2026/9*
M.Sc. in Computer Science
Research focus: Password-based authentication systems (MPC), privacy-preserving ML, and post-quantum cryptography.

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran *2017/09 – 2020/08*
M.Sc. in Applied Mathematics
Research focus: Quantum error correcting codes, Surface codes, information theoretic cryptography.

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran *2018/09 – 2020/08*
Minor in Physics.

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran *2014/09 – 2017/08*
B.Sc. in Applied Mathematics
Ranked in the top 5% among all students.

Publications & Research Projects

Publications / Preprints:

- **UpSPA: Secure and Updatable Single Password Authentication** (submitted to WPES 2026), (ePrint version, code)
- **RoUpSPA: Robust Updatable Single Password Authentication** (preprint, to be submitted to ACM TOPS)

Selected Research Projects:

- **UpSPA / RoUpSPA: Secure and robust single-password authentication.** Designed and implemented threshold-based password-authentication protocols with registration, authentication, secret-update, and password-update phases. Extended UpSPA with PBB-based synchronization, quorum-based repair, and blockchain anchoring; compared Layer-1 anchoring with Layer-2/rollup-style batching for freshness, rollback resistance, and lower update cost. Also developing the UpSPA extension, an open-source password-manager/browser-extension prototype that avoids a local password vault by distributing protected credential state across storage providers.
- **FLDP / SpectraDP: Frequency-domain differential privacy for federated learning.** Working on spectral masking for differentially private federated learning, comparing frequency-domain selection with random DP baselines under accuracy, privacy, communication, retained-energy, and reconstruction-error metrics.
- **CURE**: Contributed code-level debugging support for a homomorphic-encryption-based split-learning system. Helped investigate Go/Lattigo implementation issues, especially around parallel encrypted experiments; this was implementation support, not paper authorship.
- **QuasarSSO: A post-quantum password-protected SSO system.** Designing a password-protected SSO direction that combines post-quantum cryptographic primitives with practical authentication-system constraints.
- **AEGIS** Studying security-model and implementation issues for recent PQC-IoT handshake protocols, with emphasis on mechanizable analysis and performance-aware design.

Professional Experience

BTL Coin Crypto Exchange, Istanbul, Turkey /Remote *2021/05 – 2023/10*
Developed personalized market screeners and automated crypto asset management bots using `Golang` and `Docker`.

Institute for Research in Fundamental Sciences (IPM), Tehran, Iran *2020/10 – 2021/03*
Optimized quantum kernels for neural networks using Qiskit and PyQuil.

Paid University Project, Tehran, Iran *2019/05 – 2020/07*
Developed a data warehouse and integrated data using Excel and Tableau.

Honors & Awards

- Awarded double major recognition from the Exceptional Talent Center at the university.
- Admitted to the Master of Science program without a national entrance exam, recognized by the Exceptional Talent Center.
- Ranked in the top 5% among all students in the B.Sc. program.

Skills

- **Languages:** Persian (Native), English (Working Professional Proficiency), Turkish (Elementary).
- **Programming:** Rust, Python, Go (Golang), MATLAB.
- **Typesetting:** LaTeX.
- **Specific libraries:** Qiskit, PyQuil, Lattigo.
- **Optimization:** `scipy.optimize`, CVXPY.

Academic Services

- **External reviewer for: IEEE TIFS (2024–present), ACM ASIACCS (2024, 2026), ESORICS (2025, 2026), ACNS (2026), Royal Society Open Science (2024), PeerJ (2026), JISA (2025), and Turkish Journal of Electrical Engineering & Computer Sciences (2025)**
- Mentored two interns during Summer 2025; mentoring four interns from Fall 2025 through Summer 2026.

Teaching Experience

Amirkabir University of Technology

• Time Series Analysis	Fall 2019
• Multivariable Calculus	Spring 2019
• Introduction to C Programming	Fall 2018
• Partial Differential Equations	Fall 2018
• Foundation of Matrix and Linear Algebra	Spring 2018
• One Variable Calculus	Fall 2017
• Introduction to Probability	Spring 2017
• General Physics (Electricity and Magnetism)	Spring 2017
• Foundation of Mathematical Analysis	Spring 2017
• Introduction to Set Theory and Logic	Fall 2016–Spring 2018

Koç University

• Introduction to Python Programming	Fall 2023
• Discrete Structures	Spring 2024
• Data Structures & Algorithms	Fall 2024
• Computer Networks & Security	Spring 2025, Spring 2026
• Modern Cryptography	Fall 2025

Conferences & Workshops

- 5th and 6th Frontiers in Mathematical Sciences Conferences, IPM, Tehran, July 2017 and July 2018.
- Workshop on Mathematics of Information-Theoretic Cryptography, IMS, Singapore, September 2016.

References

- Prof. Alptekin Küpçü (link, email)
- Dr. Erfan Salavati (link, email)
- Prof. Masoud Pourmahdian (link, email)